Project Design Phase-I
Solution Architecture

Date	19 September 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - xxx
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

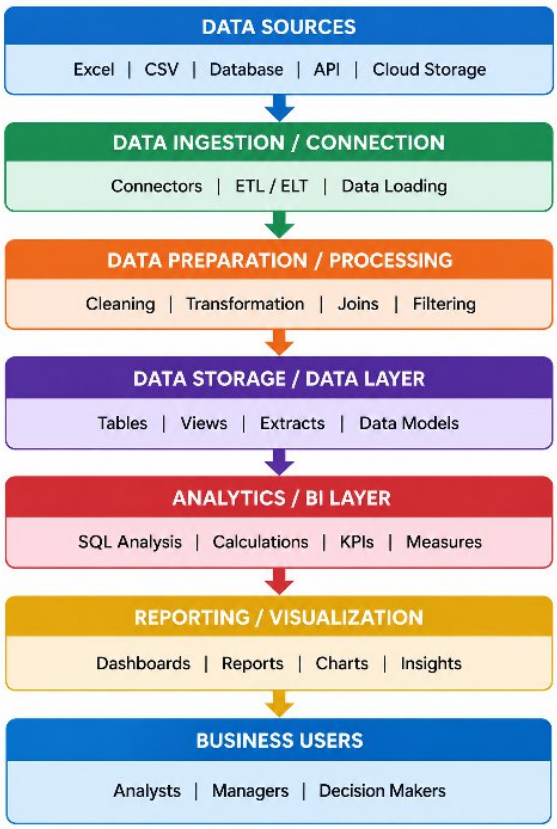
Find the best tech solution to solve existing business problems.

Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.

Define features, development phases, and solution requirements.

Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technology Used

Project Details

Date	1 November 2026
Team ID	To be provided
Project Name	Global Olympic Sports Performance Analysis Using Tableau
Prepared By	Aradhana Nagar

Proposed Solution Architecture

Data Source (athlete_events.csv + optional investment/region/host lookup files) → Data Validation & Preparation → Tableau Data Source / Relationships → Calculated Fields & Filters → 9 Worksheets → 3 Dashboards → 4-Scene Tableau Story → Publish to Tableau Public/Server → Optional Flask Web Interface with embedded Tableau views.

Architecture Layers

Component	Description
1. Data input	CSV Olympic athlete-event records; optional country-region mapping, host-city/host-NOC mapping, and country-year sports investment table.
2. Data preparation	Check schema and data types; handle null medals; validate year/season; standardize NOC codes; identify team-event duplicate medal records; document metric definitions.
3. Analysis layer	Create medal count, distinct medal key, medals per athlete, medal efficiency, and host-country medal calculations. Use distinct counts where team-event duplication is possible.
4. Visualization layer	Nine worksheets grouped into three dashboards: national performance; investment and regional efficiency; historical trends and host performance.
5. Interaction layer	Year, season, NOC, sport, medal, and region filters; dashboard actions and tooltips.
6. Presentation layer	Four story points for project overview and the three stakeholder scenarios; optional Flask page embeds

	published Tableau dashboards.
7. Validation & governance	Reconcile totals against source data; test filters and null handling; record data coverage and limitations; avoid presenting illustrative values as measured results.

Dashboard and Visualization Mapping

Dashboard	Included Visualizations
Dashboard 1 – National Olympic Performance	1. Average medal-winning athlete records per athlete by sport (bar); 2. distinct medal count by NOC (bar); 3. medal distribution by sport (treemap).
Dashboard 2 – Investment and Regional Efficiency	4. Sports investment vs medal performance (scatter/ bubble; investment data required); 5. regional medal efficiency (regional mapping required); 6. country-wise medal efficiency (bar).
Dashboard 3 – Olympic Trends and Host Performance	7. unique medal trend by year and season (line); 8. host nation medal comparison (host lookup required); 9. regional medal distribution by sport (donut; regional mapping required).

Important Data Limitations

The uploaded athlete_events.csv contains 271,116 athlete-event rows and 15 fields, with Olympic records from 1896 through 2016. It cannot substantiate claims for 2020, 2024, or 2026. It also contains no sports-investment or region column. Add verified supplementary sources before completing investment and regional analyses. Host-country analysis requires a separate year/season-to-host-NOC lookup. Medal metrics must distinguish athlete medal records from unique event medals.